

WORK BREAKDOWN STRUCTURE

Warehouse Inventory & Pallet Location Tracker

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Problem Statement #27 — Warehouse Inventory & Pallet Location Tracker

1. Project Objective

Break the Warehouse Inventory & Pallet Location Tracker project into manageable work packages covering requirements engineering, planning, system design, implementation-oriented activities, AI-assisted development evidence, testing and final documentation.

2. WBS Hierarchy

WBS ID	Work Package / Task	Description
1.0	Warehouse Inventory & Pallet Location Tracker	Overall project
1.1	Requirements Engineering	Identify and document project requirements
1.1.1	Identify stakeholders and actors	Warehouse Operator, Logistics Supervisor, Barcode/RFID Scanner
1.1.2	Define functional requirements	FR-001 to FR-005
1.1.3	Define non-functional requirements	NFR-001 to NFR-002
1.1.4	Create Requirements Traceability Matrix	Trace requirements to epics, tests and defects
1.2	Project Planning & Management	Organize project work and tracking
1.2.1	Create GitHub repository	Project repository and document organization
1.2.2	Create Jira project	Project tracking environment
1.2.3	Create epics	Seven project epics
1.2.4	Create backlog and tasks	Prioritized backlog, tasks and subtasks
1.2.5	Configure Kanban board	Track work using To-do, In-progress and Done
1.2.6	Plan Scrum sprints	Sprint 1 and Sprint 2
1.2.7	Track sprint progress	Active board and burndown evidence
1.3	System Analysis & Design	Define system behaviour and structure
1.3.1	Identify actors	Warehouse Operator, Logistics Supervisor, Scanner
1.3.2	Create UML use-case diagram	Show actor-to-function interactions
1.3.3	Define use-case flows	Preconditions, postconditions, main and alternate flows
1.3.4	Prepare SRS	Consolidate requirements and use-case specifications
1.4	System Functionality	Functional areas derived from requirements
1.4.1	Rack capacity validation	Validate placement against maximum rack capacity
1.4.2	Pallet inventory recording	Record SKU, quantity and 3D shelf location
1.4.3	Pallet location search	Search using SKU and display 3D location
1.4.4	Stock movement logging	Record movement details and timestamps
1.4.5	Rack occupancy monitoring	Display occupancy and pallet weight information
1.4.6	High-speed SKU lookup	Support lookup requirement across 100,000 bins
1.4.7	Stock movement reliability	Preserve confirmed movement records
1.5	AI-Assisted Development	GitHub Copilot evidence
1.5.1	Use GitHub Copilot	Generate or modify project code
1.5.2	Review generated code	Check generated output before use
1.5.3	Capture Copilot evidence	Screenshots/repository evidence

WBS ID	Work Package / Task	Description
1.6	Software Testing	Verification and defect handling
1.6.1	Prepare test cases	Map tests to requirements
1.6.2	Execute test cases	Run requirement-level tests
1.6.3	Identify defects	Document observed bugs
1.6.4	Apply fixes / patches	Correct identified defects
1.6.5	Retest	Verify corrected behaviour
1.7	Documentation & Submission	Prepare final submission artifacts
1.7.1	Organize deliverable folders	Six required submission folders
1.7.2	Collect GitHub/Jira evidence	Screenshots and project-management artifacts
1.7.3	Finalize README	Index and describe all deliverables
1.7.4	Final repository review	Check completeness and links

3. Project Planning Alignment

The WBS is aligned with the project's existing Kanban and Scrum planning artifacts. The Scrum documentation records Sprint 1 as a one-week sprint containing 21 story points and Sprint 2 as containing the remaining 19 story points. The WBS provides the hierarchical work decomposition, while the Jira boards and sprints provide the scheduling and progress view.

4. Completion Criteria

- All functional and non-functional requirements are documented and traced.
- The UML/use-case design documentation is included.
- Kanban and Scrum project-management evidence is included.
- SRS and WBS are finalized.
- GitHub Copilot evidence is captured.
- Testing, bug fixing and retesting evidence is included as required by the submission.
- The final GitHub repository contains the required folders and README.